SD1 Viva Prep and Practice Questions

Software Development I (4COSC006C) · December 2023 · typed up from chat screenshots, a Colab notebook and quiz screenshots

1. Coursework Viva - Questions Students Were Asked

Collected from classmates after their SD1 vivas (December 2023). Not every student was asked every question.

Coding tasks (live, in the viva)

- Print the letters of the word "Westminster" one per line (a for loop).
- Print a shape made of * characters using loops.
- Write a loop that prints a multiplication table, e.g. 2*1=2, 2*2=4, ...
- Print the last letter of "hello world".
- Create a list containing strings and ints, print it using a for loop, then append another element to the end of the list.
- Explain your coursework code (the first-year progression / marks calculation) and the Python behind it.
- Explain and run a small program that uses a list, a dictionary or a function.

Theory questions

- What is the difference between = and ==?
- What is the difference between print and return?
- What is the difference between a list and a dictionary?
- What is a user-defined function?
- Give some examples of errors. Can a syntax error be handled with an exception?

2. Viva Practice Questions (VivaQuestions.ipynb)

Question 1

Print the prime numbers from 1 to 100.

```
for i in range(1, 101):
    for j in range(2, i + 1):
        if i % j == 0:
            if i == j:
                print(i, ' ', sep='', end='')
            break
```

Question 2

Print the elements that are unique to each of two lists.

```
List1 = ['red', 'green', 'yellow', 'black']
List2 = ['red', 'purple', 'pink', 'green']

for item in List1:
    if item not in List2:
        print(item, end=' ')
for item2 in List2:
    if item2 not in List1:
        print(item2, end=' ')

# Output: yellow black purple pink
```

Question 3

Generate 100 random numbers and print the first 20 odd numbers.

```
import random

random_list = []
for i in range(0, 100):
    random_number = random.randint(0, 100)
    random_list.append(random_number)

odd_numbers = []
for item in random_list:
    if len(odd_numbers) == 20:
        break
    elif item % 2 != 0:
        odd_numbers.append(item)

for item in odd_numbers:
    print(item, ' ', sep='', end='')
```

Question 4

Print the even numbers between 1 and 100.

```
for n in range(1, 101):
    if n % 2 == 0:
        print(n, ' ', end='', sep='')
```

Question 5

Print the colours common to List1 and List2.

```
List1 = ['red', 'green', 'yellow', 'black']
List2 = ['red', 'purple', 'pink', 'green']

for item in List1:
    if item in List2:
        print(item, ' ', end='', sep='')

# Output: red green
```

Question 6

Print the odd numbers between 1 and 100. (Same as Q4 with n % 2 != 0.)

Note: the notebook's first lines (from traitlets.traitlets import ...) were an accidental auto-import and are not needed, so they are left out above.

3. Practice Quiz Question

Question 1

```
if x > 0:
    if y > 0:
        print("x > 0 and y > 0")
elif z > 0:
    print("x < 0 and z > 0")
```

Suppose x = 1, y = -1 and z = 1. What will be displayed by the following statement?

A. x < 0 and z < 0

B. nothing displayed

C. x < 0 and z > 0

D. x > 0 and y > 0

Answer: B - nothing displayed. x > 0 is true, so the elif is skipped; inside the if, y > 0 is false and there is no else.

4. Revision Lecture Topics (Week 11)

1. Basics: arithmetic operators, precedence levels
2. Conditions
3. Loops, nested loops, break and continue
4. Lists, strings, subscript operations, slices
5. Mutable / immutable
6. Lists: built-in operations, list comprehension